Phase 2 Capstone Report

Student Grade Analyser with Pattern Matching

BACHELOR OF TECHNOLOGY (CSE/ AI&ML)

Semester-VIII

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1. Project Description

The Student Grade Analyser is an F# console-based application that analyses student examination marks and
presents academic results. It assigns grades using pattern matching, calculates the class average, identifies the
highest-scoring student, and displays pass/fail statistics through a simple menu-driven interface.

2. Objectives

- Assign grades (A+, A, B, C, D or F) according to predefined mark ranges using pattern matching.
- Store student names and marks using F# records and an immutable list.
- Display each student's name, marks, grade and Pass/Fail result.
- Calculate the class average and identify the topper.
- Count the number of students who passed and failed.
- Provide a menu-driven interface with an option to exit and handling for invalid choices

3. Key Features

Feature	Description
Grade Assignment	Assigns grades based on the student's marks using F# pattern matching.
Student details	Displays student names, marks, grades and pass/fail results in a formatted table.
Class average	Calculates the average marks of all students.
Topper identification	Finds the student with the highest marks.
Pass/fail count	Displays the number of students who passed and failed.
Interactive menu	Allows users to select an operation or exit the application.

4. Technologies Used

Technology	Purpose
F#	Main programming language used to develop the application.
FSharp.Core	Provides records, lists, pattern matching, higher-order functions and the pipeline operator.
System (.NET)	Used for console input and output.
.NET SDK 8.0 or later	Used to build and run the F# console application.
Visual Studio Code with Ionide	Used for writing and editing F# code.
GitHub	Used to host and share the project source code.

5. Implementation

The application defines a Student record containing the student's name and marks. Five sample student records are stored in an immutable list. The getGrade function uses a match expression with guards to assign a grade according to the marks. The showStudents function uses List.iter to display each student's details andPass/Fail status.

The class average is calculated using List.averageBy, while List.maxBy identifies the topper. List.filter separates students into passed and failed groups. A while loop displays the menu repeatedly, and a match expression processes the user's selection and calls the corresponding operation.